orbits in gravitational fields

Rockets and satellites that orbit a planet or star have both KE and GPE when in orbit.

$$\therefore \text{Total energy of an orbiting satellite} = E = E_p + E_k$$

To find the expression for E from first principals, express E_p and E_k separately.
Let M be the mass of the planet and m be the mass of the orbiting object (satellite).

Evaluating E_p:

$$\begin{aligned} \text{From A.2,} \quad \Delta E_p &= mg\Delta h \\ E_p &= mgr \\ &= m \times (-G\frac{M}{r^2}) \times r \\ &= -G\frac{Mm}{r} \\ &\equiv -G\frac{m_1 m_2}{r} \quad \text{as given in formula book} \end{aligned}$$

Before evaluating E_k, express centripetal force:

$$\begin{aligned} F_{\text{cent}} &= ma_{\text{cent}} \\ a_{\text{cent}} &= \frac{v^2}{r} \\ \therefore F_{\text{cent}} &= \frac{mv^2}{r} \end{aligned}$$

Now express E_k:

$$\begin{aligned} \text{grav. force} &= \text{centripetal force} \\ g &= F_{\text{cent}} \\ \therefore G\frac{Mm}{r^2} &= \frac{mv^2}{r} \\ \frac{1}{2} \times G\frac{Mm}{r^2} &= \frac{1}{2} \times \frac{mv^2}{r} \\ G\frac{Mm}{2r^2} &= \frac{mv^2}{2r} \\ G\frac{Mm}{2r} &= \frac{1}{2}mv^2 \\ \therefore E_k &= G\frac{Mm}{2r} \end{aligned}$$

Since $E = E_p + E_k$:

$$\begin{aligned} E &= (-G\frac{Mm}{r}) + (G\frac{Mm}{2r}) \\ &= -G\frac{Mm}{2r} \end{aligned}$$

If the above is mathematically unclear, let variable $u = G\frac{Mm}{r}$ and notice $E = -u + \frac{u}{2}$.

Yes, this means that E is negative.
$\implies$ the satellite is bound in its orbit and cannot escape from the planet.

Since $E_k = -\frac{E_p}{2}$, when the satellite moves to a lower orbit, r decreases
$\implies$ E_k increases, but E_p decreases 2 times more (increase in the negative direction).

The expression for E_k can be used to derive $v_{orbital}$ in the formula book, which is a slightly different way to the derivation here.

The **orbital speed** of the satellite increases as the orbital radius decreases even though it has lost energy (overall). This is because the total energy becomes more negative — that is, more tightly bound to the system.

erm, acktually. what about drag?

A satellite in a low Earth orbit is **subject to drag** caused by **collisions with the ions and molecules in the atmosphere** $\implies$ $\exists$ a drag force in a direction at a **tangent** to the direction of motion of the satellite, in the **opposite direction** to the **linear velocity** of the satellite.

This atmospheric drag removes energy from the Earth-satellite system. Recall,

$$E = -G\frac{Mm}{2r}$$

$\therefore$ E decreases $\implies$ r decreases $\implies$ E_k increases $\implies$ v_{orbital} increases

This whole discussion of drag is true $\iff$ the drag force is small.

If the drag force is large, the **deceleration of the satellite** will be so great and the E_k will be lost so quickly that the satellite **cannot be treated as being in orbit.** It now behaves like an intercontinental ballistic missile or an unidentified flying object — it's a **projectile** subject to Earth's gravity.

Escape speed of a system: the minimum speed required for an object to leave a gravitational field and (just) reach infinity $\implies$ PE = 0.

Recall that $E_p = -G\frac{Mm}{r}$.

At escape speed, initial E_k must be equal to $|E_p|$ at the point where the satellite is in the field, at distance r from the centre of the sphere that is causing the gravitational field.

$$E_k = |E_p|$$
$$\frac{1}{2}m{v_{\text{esc}}}^2 = G\frac{Mm}{r}$$
$$\therefore v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$$

With regards to $v_{\text{orbital}} \iff$ **at the planet surface**,

$$v_{\text{esc}} = \sqrt{2} \times v_{\text{orbital}}$$

The relationship is true if, and only if, the body is at the planet surface as we are dealing with **initial** kinetic energy.

Geosynchronous orbit: an orbit with an orbital period that matches the Earth's rotation on its axis.

Geostationary orbit: An orbit that is geosynchronous and also **positioned above the equator.** It remains apparently fixed in position when viewed from Earth.
